

SIMATS ENGINEERING

ASSESSMENT TOOL 3

MCQ Test

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

Total Marks: 20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
- a) Slow Start
 - b) Fast Retransmit
 - c) Fast Recovery
 - d) Tahoe Reset

Answer: c) Fast Recovery

2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
- a) 9 MSS
 - b) 16 MSS
 - c) 12 MSS
 - d) 8 MSS

Answer: b) 16 MSS

3. Which field in the TCP header is used to implement flow control at the receiver side?
- a) Sequence number
 - b) Acknowledgement number
 - c) Window size
 - d) Urgent pointer

Answer: c) Window size

4. Nagle's algorithm causes increased latency primarily because it:
- a) Disables ACK piggybacking
 - b) Buffers small segments until ACK arrives
 - c) Reduces CWND on every packet
 - d) Increases SYN backlog

Answer: b) Buffers small segments until ACK arrives

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:
- a) MTU size
 - b) CWND growth rate relative to RTT
 - c) DNS resolution time
 - d) UDP competition

Answer: b) CWND growth rate relative to RTT

6. UDP is preferred over TCP for DNS queries primarily because:
- a) UDP supports fragmentation
 - b) DNS responses are small and connection overhead is unnecessary
 - c) TCP cannot use port 53
 - d) UDP has better error correction

Answer: b) DNS responses are small and connection overhead is unnecessary

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:
- a) Increase UDP buffer size
 - b) Switch to TCP
 - c) Implement Forward Error Correction (FEC)
 - d) Reduce frame rate to 1 fps

Answer: c) Implement Forward Error Correction (FEC)

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?
- a) 1480 bytes
 - b) 1472 bytes
 - c) 1460 bytes
 - d) 1500 bytes

Answer: b) 1472 bytes

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:
- a) Eliminating IP header overhead
 - b) 0-RTT or 1-RTT connection setup
 - c) Removing encryption
 - d) Native multicast support

Answer: b) 0-RTT or 1-RTT connection setup

10. HTTP/1.1 persistent connections reduce latency by:
- a) Enabling UDP fallback
 - b) Reusing the same TCP connection for multiple requests
 - c) Compressing HTTP headers
 - d) Sending requests without waiting for DNS

Answer: b) Reusing the same TCP connection for multiple requests

11. Head-of-line blocking in HTTP/1.1 occurs when:

- a) The DNS server is slow
- b) A response blocks subsequent responses on the same connection
- c) TCP CWND drops to zero
- d) The server rejects pipelined requests

Answer: b) A response blocks subsequent responses on the same connection

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

- a) Application layer
- b) DNS layer
- c) TCP layer
- d) IP layer

Answer: c) TCP layer

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

- a) 301
- b) 302
- c) 304
- d) 307

Answer: a) 301

14. In HTTP caching, the Cache-Control: no-cache directive means:

- a) Never cache this resource
- b) Validate with the server before using a cached copy
- c) Cache for 0 seconds
- d) Disable all caching headers

Answer: b) Validate with the server before using a cached copy

15. Which DNS record type maps a domain name to an IPv6 address?

- a) A
- b) CNAME
- c) AAAA
- d) MX

Answer: c) AAAA

16. A DNS resolver switches from UDP to TCP when:

- a) The query type is MX
- b) The response is truncated (TC bit set)
- c) TTL exceeds 3600 s
- d) The authoritative server is unreachable

Answer: b) The response is truncated (TC bit set)

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

- a) ARP poisoning
- b) DNS cache poisoning
- c) SYN flooding
- d) BGP hijacking

Answer: b) DNS cache poisoning

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

- a) Cache hit ratio
- b) Recursive resolver query load
- c) DNSSEC signature validity
- d) Zone transfer frequency

Answer: b) Recursive resolver query load

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

- a) HTTP → TCP → DNS
- b) DNS → TCP handshake → HTTP GET
- c) TCP → DNS → HTTP
- d) DNS → HTTP → TCP

Answer: b) DNS → TCP handshake → HTTP GET

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

- a) TCP for game state, HTTP for assets
- b) UDP with application-layer sequencing for game state, HTTP/3 for assets
- c) HTTP/2 for all traffic
- d) TCP with Nagle enabled for all traffic

Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Code of Conduct:

I LOKESH S (192572151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

LOKESH S

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions